

MFE5200 – Computational Methods in Financial Engineering
Summer Term, 2025-26

Assignment 2

Due date: 11:00 pm, 2-July-2026

Important notes:

1. The assignment must be submitted via Blackboard.
 2. The program codes of the programming problems (if any) must also be submitted.
 3. Total: **10 Problems** (Full Mark: 100).
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Problem 1

Let $S(t)$ be the price of a non-dividend paying asset at time t . The dynamics of $S(t)$ under the risk-neutral probability measure Q is governed by

$$\frac{dS(t)}{S(t)} = rdt + \sigma dW(t),$$

where r is the continuously compounded risk-free interest rate and $W(t)$ is a standard Brownian motion under Q .

Derive the pricing formula of a T -year Asian option with the payoff of

$$\max(\bar{S}_G - K, 0),$$

where

$$\bar{S}_G = \left(\prod_{i=1}^n S(t_i) \right)^{\frac{1}{n}}$$

over the set of monitoring dates $0 < t_1 < t_2 < \dots < t_n < T$.

Note:

If $Y \sim \text{LN}(\mu, \beta^2)$, then, for some fixed constant C , we have

$$E[\max(Y - C, 0)] = E[Y]N(d_1) - CN(d_2),$$

where

$$d_1 = \frac{\log\left(\frac{E(Y)}{C}\right) + \frac{1}{2}\beta^2}{\beta}, \quad d_2 = d_1 - \beta.$$

Problem 2

The dynamics of $S_1(t)$ and $S_2(t)$ are governed by

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i dX_i(t), \quad i = 1, 2,$$

where each $X_i(t)$ is a standard one-dimensional Brownian motion and $dX_1(t)dX_2(t) = \rho dt$.

Suppose that $S_1(0) = 5$, $S_2(0) = 6$, $\mu_1 = 0.03$, $\mu_2 = 0.05$, $\sigma_1 = 0.3$, $\sigma_2 = 0.2$ and $\rho = -0.2$.

Find $\text{Cov}(S_1(0.5), S_2(0.5))$.

Problem 3

Show that optimal value of b in the control variate method is given

$$\frac{\text{Cov}(X, Y)}{\text{Var}(X)}.$$

Problem 4

Suppose that X is a Bernoulli random variable with parameter p . Show that if $2p > 1$, then the antithetic variable of X is

$$Y = 1 - XZ,$$

where Z is a Bernoulli random variable with parameter $\frac{1-p}{p}$, independent of X .

Note:

To show Y is the antithetic variable of X , you need to show that X and Y follow the same distribution and $\text{Cov}(X, Y) < 0$.

Problem 5

Use the antithetic variates method to estimate $P(Z \geq 0.8)$, where Z is a standard normal random variable.

- (a) Write down the antithetic variates estimator in terms of $Z_1, Z_2, \dots, Z_n$ where are the i.i.d. copies from the standard normal distribution.
- (b) Compare to the plain Monte Carlo simulation with $2n$ independent replications, determine the percentage of variance reduction achieved by the antithetic variates method using n antithetic pairs.

Problem 6

Let U be a random variable uniformly distributed on $[0, 1]$.

Consider the problem of estimating $E[e^U]$ by stratified sampling. Stratify U by partitioning the interval $[0, 1]$ into $k = 2$ strata of equal length.

Let $V_{11}, V_{12}, \dots, V_{1n}$ are independent and uniformly distributed between 0 and 0.5.

Let $V_{21}, V_{22}, \dots, V_{2n}$ are independent and uniformly distributed between 0.5 and 1.

- (a) Write down the stratified estimator with the proportional allocation in terms of $V_{11}, V_{12}, \dots, V_{1n}, V_{21}, V_{22}, \dots, V_{2n}$.
- (b) Evaluate the magnitude of variance reduction from the estimator in (a).
- (c) Find the optimal allocation and write down the corresponding estimator.

Problem 7

Let X be a binomial random variable with parameters (m, p) such that

$$\Pr(X = k) = \frac{m!}{(m-k)!k!} p^k (1-p)^{m-k}.$$

Write down the importance sampling estimates for the following:

$$\Pr(X \geq m\alpha),$$

where $\alpha \in (0, 1)$ is a given constant.

The alternative sampling distribution is binomial with parameters $(m, \bar{p})$.

Problem 8

Let X be a standard normal random variable. Assume that we wish to use importance sampling to estimate, for a given positive constant b ,

$$E\left(\mathbf{1}_{\{X \geq b\}}\right)$$

Taking $g(x)$ be the probability density function (pdf) of $N(\theta, 1)$ for some $\theta \in \mathcal{R}$.

- (a) Let $\theta = x^*$, where x^* maximizes $\mathbf{1}_{\{x \geq b\}}f(x)$ where $f(x)$ is the pdf of $N(0, 1)$. Determine θ .
- (b) Let $g(x)$ be the pdf of $N(x^*, 1)$. Using the importance sampling method with alternative pdf of $g(x)$, write down the importance sampling estimator of $E\left(\mathbf{1}_{\{X \geq b\}}\right)$ in terms of $X_1, X_2, \dots, X_n$ which are the i.i.d. copies from $N(x^*, 1)$.

Problem 9

Determine the exponential tilt family of the random variable X .

- (a) X is Bernoulli with parameter p .
- (b) X is exponentially distributed with mean $\frac{1}{\lambda}$.

Problem 10

Assume that X is a random variable with density f , and $\{f_\theta\}$ is an exponential tilt family.

Let $H(\theta) = \log E[e^{\theta X}]$. Show that the updating rule for the tilting parameter θ in the general iterative cross-entropy scheme is determined by

$$H'(\hat{\theta}^{j+1}) = \frac{\sum_{k=1}^N h(Y_k) \ell_{\hat{\theta}^j}(Y_k) Y_k}{\sum_{k=1}^N h(Y_k) \ell_{\hat{\theta}^j}(Y_k)},$$

where $Y_1, \dots, Y_N$ are i.i.d. pilot samples from the density $f_{\hat{\theta}^j}$.

End

NORMAL DISTRIBUTION TABLE

Entries represent the area under the standardized normal distribution from $-\infty$ to z , $\Pr(Z < z)$

The value of z to the first decimal is given in the left column. The second decimal place is given in the top row.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Values of z for selected values of $\Pr(Z < z)$							
z	0.842	1.036	1.282	1.645	1.960	2.326	2.576
$\Pr(Z < z)$	0.800	0.850	0.900	0.950	0.975	0.990	0.995